

1 Summary

Model (Run-name): model1 (test)

Iteration: 12

TAC: 154560.69361165454

Took: 2.109113931655884s

Total Time: 15.034610748291016s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

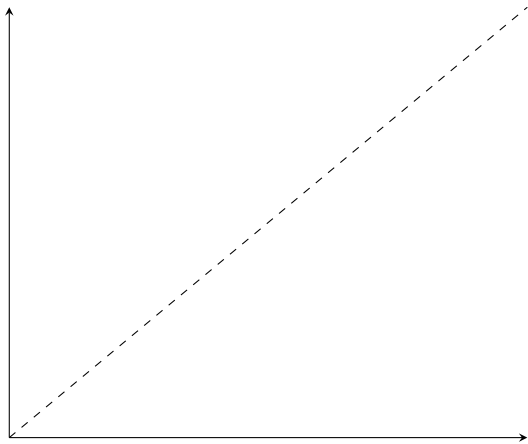
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

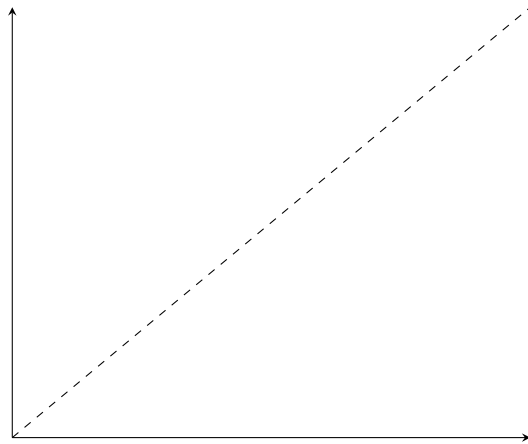
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05265288183144592	$b_{1,1,1}^{H,out}$	880.8999765673402
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017250443844034985	$b_{1,2,2}^{H,out}$	3799.381245171851
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.030312524151545932	$b_{2,1,2}^{H,out}$	9349.940513941305
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014457324633612916	$b_{1,1,1}^{C,in}$	6043.622868684205
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009616643394353735	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022640356318941436	$b_{2,1,2}^{C,in}$	4228.205677365226
$A_{1,1,1}^\beta$	69.42777867315647	$q_{1,1,1}$	666.6270092942034	$b_{1,1,1}^{C,out}$	6710.249877978407
$A_{1,2,2}^\beta$	67.27673099173637	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	147.07009017600456	$q_{2,1,2}$	2450.0594860586953	$b_{2,1,2}^{C,out}$	6678.265163423919
$A_{cu,1}^\beta$	4.983938626310074	$q_{cu,1}$	183.37299070579684		
$A_{cu,2}^\beta$	37.443498367229196	$q_{cu,2}$	1949.9405139413047		
$A_{hu,1}^\beta$	13.104034300095861	$q_{hu,1}$	483.3135046471017		
$A_{1,1,1}$	69.42777867315647	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	67.27673099173637	$t_{1,2}^{(H)}$	583.3372990705797		
$A_{2,1,2}$	147.07009017600456	$t_{1,3}^{(H)}$	388.33729907057966		
$A_{cu,1}$	4.983938626310074	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.443498367229196	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.104034300095861	$t_{2,3}^{(H)}$	467.4970256970652		
$dt_{1,1,1}$	32.22090030980677	$t_{1,1}^{(C)}$	617.7790996901932		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	573.3372990705797		
$dt_{1,2,2}$	83.33729907057966	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	38.33729907057969	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	16.66270092942034	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	57.49702569706527	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	68.33729907057969	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	147.49702569706523	$t_{1,2,2}^H$	385.49645730629237		
$dt_{hu,1}$	62.22090030980678	$t_{2,1,2}^H$	467.4970256970652		
$f_{1,1,1}^H$	2.380810747479298	$t_{1,1,1}^C$	636.9614739994557		
$f_{1,2,2}^H$	9.855813647991816	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	648.3341506855953		
$f_{1,1,1}^C$	10.534781383001103	$b_{1,1,1}^{H,in}$	1547.5269858615436		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5749.381245171851		
$f_{2,1,2}^C$	10.312696774061525	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

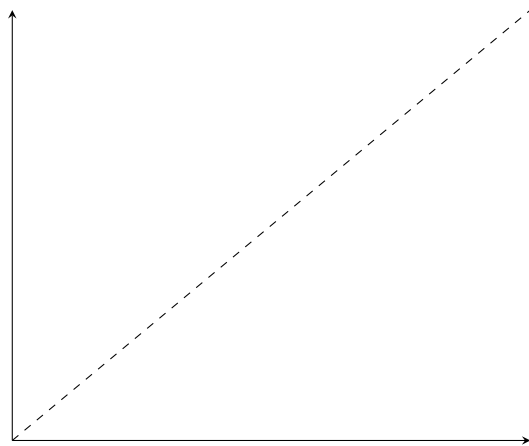
- $(1, 1, 1)$
 - Value: 0.052653
 - Error: 0.000001610589110
 - Relative error: 0.000030587876472
- $(1, 2, 2)$
 - Value: 0.017250
 - Error: 0.000004510123732
 - Relative error: 0.000261381383018
- $(2, 1, 2)$
 - Value: 0.030313
 - Error: 0.000018832673710
 - Relative error: 0.000620897832511
- $cu, 1$
 - Value: 0.014457
 - Error: 0.000000788463252
 - Relative error: 0.000054534312106
- $cu, 2$
 - Value: 0.009617
 - Error: 0.000000664651441
 - Relative error: 0.000069109925335
- $hu, 1$
 - Value: 0.022640
 - Error: 0.000000028414881
 - Relative error: 0.000001255052942



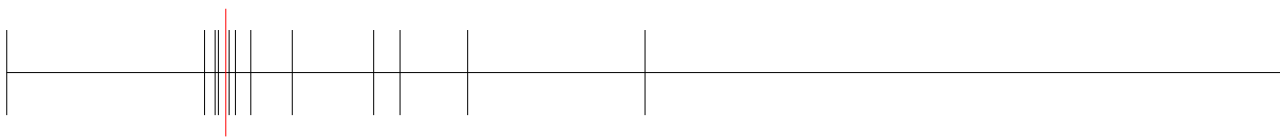
(a) *RecLMTD* - (1, 1, 1)



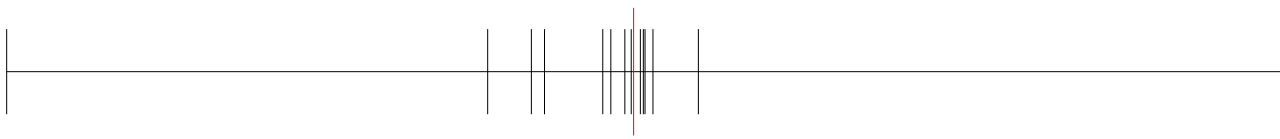
(b) *RecLMTD* - (1, 2, 2)



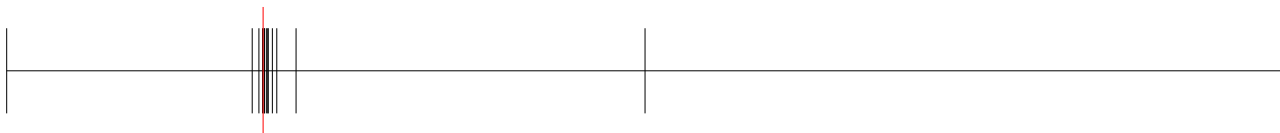
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1547.5269858615436
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5749.381245171851
 - Absolute error: 0.11753160934858897
 - Relative error: 2.0442897595970788e-05
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 880.8999765673402
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3799.381245171851
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9349.940513941305
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 6043.622868684205
 - Absolute error: 3.639764255326554
 - Relative error: 0.000602611661720984
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4228.205677365226
 - Absolute error: 9.094947017729282e-13
 - Relative error: 2.151018117783885e-16

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 6710.249877978407
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6678.265163423919
 - Absolute error: 7.808340865338323
 - Relative error: -0.0011678514841826234



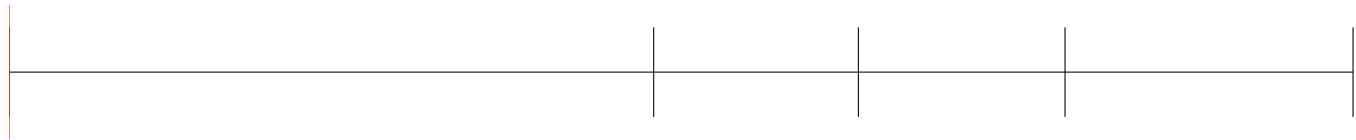
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



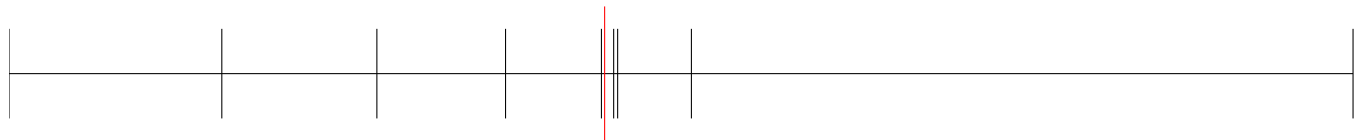
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$

5 Area Breakpoints

5.1 A

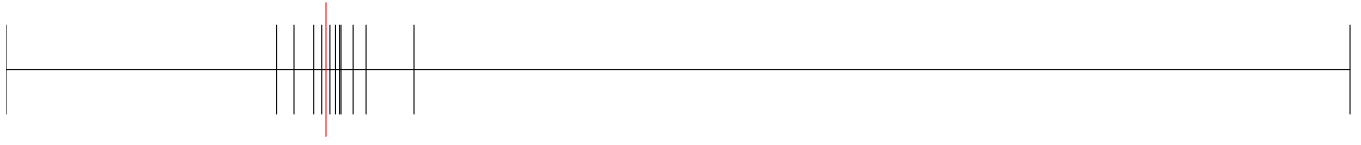
- $(1, 1, 1)$
 - Value: 69.427779
 - Error: 0.771887618879319
 - Relative error: 0.010995602396003
- $(1, 2, 2)$
 - Value: 67.276731
 - Error: 0.000000000000071
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 147.070090
 - Error: 1.464884511752274
 - Relative error: 0.009862219418906

5.2 A_{CU}

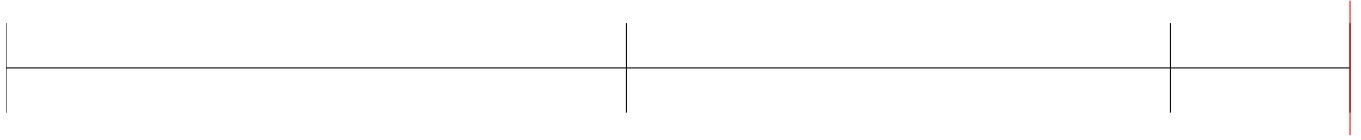
- $cu, 1$
 - Value: 4.983939
 - Error: 0.318227085030304
 - Relative error: 0.060018321258740
- $cu, 2$
 - Value: 37.443498
 - Error: 0.060266758323550
 - Relative error: 0.001606952211912

5.3 A_{HU}

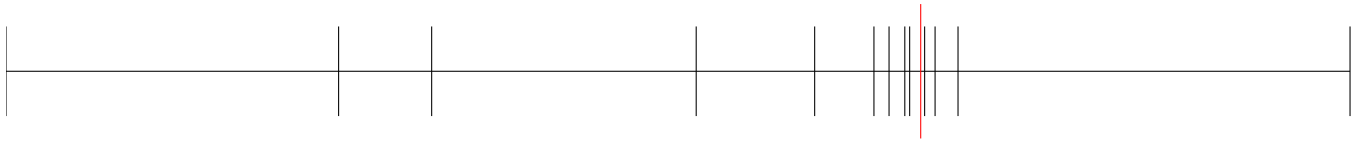
- $hu, 1$
 - Value: 13.104034
 - Error: 0.026833650664226
 - Relative error: 0.002043554985463



(a) $A^\beta - (1, 1, 1)$



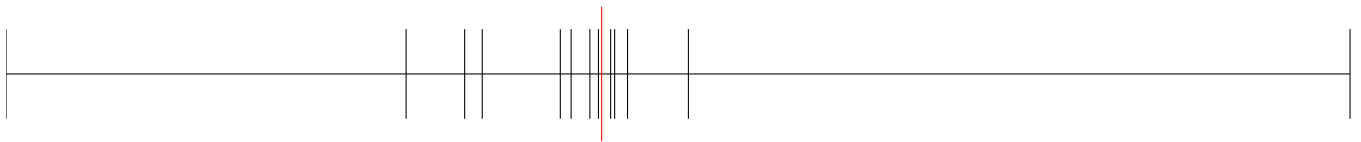
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

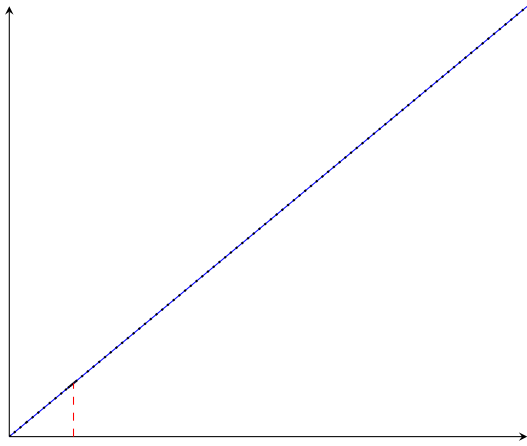


(a) $A^\beta - hu, 1$

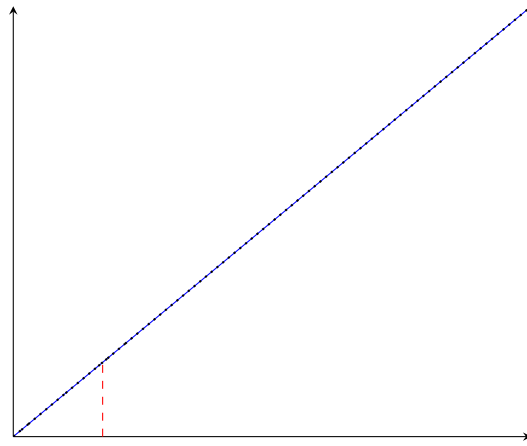
6 A^β Breakpoints

- (1, 1, 1)
 - q : 69.427779
 - $\hat{q}^\beta$: 69.427779
 - q^β : 69.427779
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.89902342903967, 70.30181758862284, 71.22933662920877, 71.6637831223933, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 67.276731
 - $\hat{q}^\beta$: 67.276731
 - q^β : 67.276731
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 69.34502840170047, 70.94081287188642, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 147.070090
 - $\hat{q}^\beta$: 147.070090
 - q^β : 147.070090
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 131.69760457629607, 132.12228195956155, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.983939
 - $\hat{q}^\beta$: 4.983939
 - q^β : 4.983939
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 4.269280092716329, 4.432280503220532, 5.0418124100090695, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.443498
 - $\hat{q}^\beta$: 37.443498
 - q^β : 37.443498

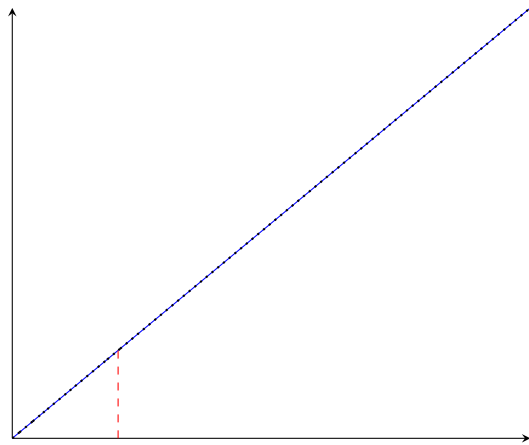
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, **37.21616748032837**, **37.47875108132638**, 37.9454585315849, 38.21652191748154, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.104034
 - $\hat{q}^\beta$: 13.104034
 - q^β : 13.104034
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, **12.917705220652834**, **13.192252946941991**, 13.22717920580809, 13.244985069124171, 13.30871858348944, 13.346967780023082, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



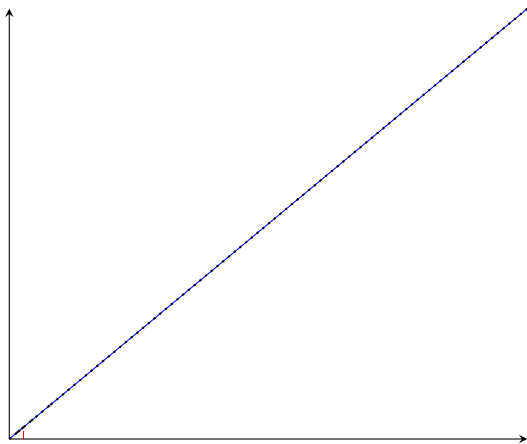
(a) $q^\beta - (1, 1, 1)$



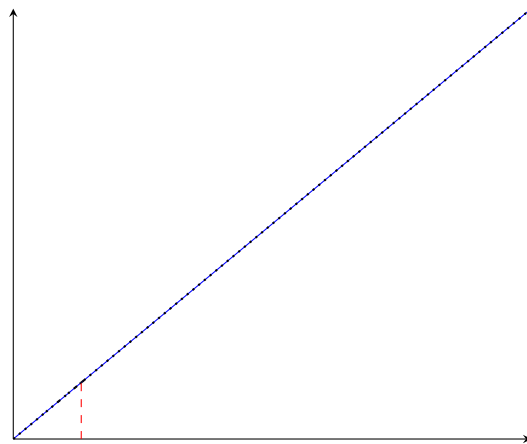
(b) $q^\beta - (1, 2, 2)$



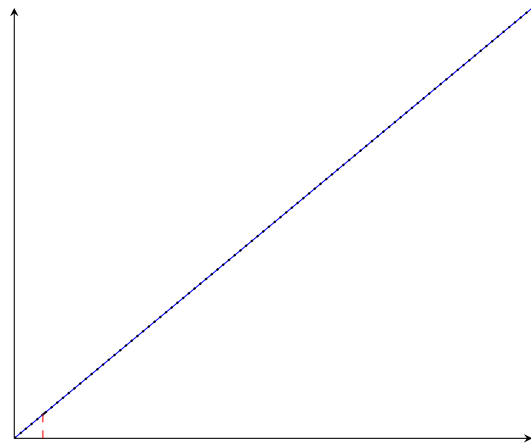
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1921.7943226347747
$z_{2,1,1}$	0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008047090550039965	$b_{2,1,1}^{C,in}$	2556.436617374491
$z_{2,2,2}$	0.0	$RecLMTD_{2,1,1}$	0.1	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.02011172883803818	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.004340277777777778	$b_{1,1,2}^{C,out}$	1921.794322634776
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	2556.436617374491
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	557.520587407834		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	379.4528007924781		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	466.6292188464114		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	370.0454403499534		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	83.99174553394612		
$f_{1,1,2}^H$	0.1441863520081838	$b_{1,2,1}^{H,in}$	4952.473014138456		
$f_{1,2,1}^H$	7.619189252520702	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	83.99174553394612		
$f_{1,1,2}^C$	4.687303225938475	$b_{1,2,1}^{H,out}$	4952.473014138457		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	4.4652186169988966	$b_{2,2,1}^{H,out}$	0.0		